Task 7 - Type Conversion and Precision

C Programming Assignment - Rural Electricity Billing System

Conversions

C performs integer promotions and usual arithmetic conversions. Mixing signed and unsigned values may change comparisons unexpectedly. Keep related calculations in consistent types.

Integer division

5 / 2 gives 2 when both operands are integers. 5.0 / 2.0 gives 2.5. Do not use floating point for final currency calculations.

Casting

A cast does not make an out-of-range conversion safe. Check the range before converting a large long long value to int.

Rounding and examples

Floating-point 0.1 + 0.2 may not equal exactly 0.3. Define a documented rounding rule if fractional paise occur. Prefer integer-paise calculations where possible.

Recommendation

Use wide integers, check overflow, keep money in paise and format currency only when displaying the result.

Important: illustrative tariff rates are used because the screenshot does not show the complete official tariff table. Replace them with your instructor's prescribed rates.